

**DEPARTMENT OF ECONOMICS UNIVERSITY OF CALABAR, CALABAR SECOND SEMESTER
EXAMINATIONS, 2024/2025 SESSION ECO206: STATISTICS FOR ECONOMISTS
TIME ALLOWED: 2 HOURS**

INSTRUCTIONS: Answer any three questions only.

- 1.** (a) Distinguish clearly between the following pairs of concepts: (i) Point estimate and interval estimate (ii) Estimator and Estimate. (b) Discuss with relevant diagrams, the criteria by which we can evaluate the quality of a statistic as a good estimator.
- 2.** (a) What is confidence interval? (b) The test scores of ten (10) are given respectively as: 80, 95, 90, 90, 95, 75, 75, 85, 90 and 80. Use the test scores to calculate the confidence interval assuming you have a confidence level or Z-value of 95% and interpret your results.
- 3.** (a) What is a hypothesis? (b) A machine fills packets with spice which are supposed to have a mean weight of 40 grams. A random sample of 36 packets is taken and the mean weight is found to be 42.4 grams with a standard deviation of 6 grams. You are required to conduct a significance test at the 5% level.
- 4.** (a) Using a diagram in the appropriate case, explain the difference between the normal probability distribution and the binomial probability distribution. (b) A study found that the residents of a city with a population of 5000 persons were expected to live for an average of 50 years (life expectancy), with a standard deviation of 20 years. How many persons in the city will live for 20 to 65 years?
- 5.** (a) If South East Asia experiences an average of 2 tsunamis every 10 years, what is the probability that in any given 10 years, there will be 3 tsunamis? (b) What is F-statistic? State the assumptions underlying the use of the F-Test. (c) Discuss decision making using the F-test.